

Flow as Insight-Driven Cognition: Implications on Consciousness

Introduction

Flow is commonly described as a state of effortless performance in which action feels automatic and one's focused attention is fully absorbed by the task. Athletes, musicians, and performers often report that during flow, actions unfold smoothly without deliberate control, and self-conscious thought appears to recede. This has led to the widespread assumption that flow reflects the suppression of higher cognition in favor of automatic, unconscious processes. This paper argues that this interpretation is incomplete or even incorrect. Vervaeke, Ferraro, and Herrera-Bennett (2018) argue that flow is not merely the execution of overlearned routines, but a state characterized by spontaneous insight and ongoing learning. In this view, flow represents a form of thought driven by insight and supported by implicit learning. Insight, in this context, refers to the sudden restructuring of a problem or task that allows for more effective action without step-by-step reasoning. Implicit learning, by contrast, involves the gradual refinement of skills through experience and feedback, often without explicit awareness of what is being learned. Together, these processes suggest that cognition in flow is both adaptive and continuously evolving, rather than fixed or purely automatic.

Flow is particularly useful as a case study for consciousness because it sits at the intersection of skilled action, subjective experience, and cognitive transformation. Unlike purely reflective states, flow involves minimal self-conscious awareness, yet cognition remains highly effective and adaptive. At the same time, it differs from automatic habits because it involves ongoing learning and sensitivity to feedback. This makes flow a unique window into the ways in which consciousness operates when it is functioning optimally, revealing processes that are often less visible in everyday cognition. This reinterpretation of flow has important implications for understanding consciousness. If flow involves active restructuring rather than passive execution, then consciousness cannot be reduced to reflective or deliberative processes alone. Instead, it suggests that consciousness functions as a dynamic system for optimizing interaction with the environment. Predictive processing provides a useful framework for developing this idea. According to this approach, perception and cognition are forms of inference in which the brain generates predictions about the world and updates them based on sensory input (Seth & Hohwy, 2021). Conscious experience is therefore not a passive reflection of reality, but an active process shaped by the minimization of prediction error.

This essay argues that flow reveals fundamental features of consciousness. Specifically, it proposes that consciousness is a self-organizing system that dynamically restructures relevance through the integration of prediction and feedback. In doing so, it addresses four central questions: the function of consciousness, its nature, its transcendental–empirical structure, and what constitutes a minded system. Flow will be used as a case study to show that consciousness is not static or purely computational, but an adaptive and continuously evolving process.

The Function of Flow: Learning Through Action

A central claim of Vervaeke et al. (2018) is that flow serves a dual function it enables skilled performance, while simultaneously improving that skill. This challenges traditional models that treat cognition as the application of fixed representations or routines. In many accounts, behavior is guided by previously acquired structures that are simply executed when needed. However, flow suggests a different picture, in which cognition remains responsive and continuously shaped by ongoing feedback

(Vervaeke et al., 2018). Rather than applying static patterns, individuals dynamically adjust their behavior in real time, refining their engagement with the task as it unfolds.

During flow, individuals engage in tasks that provide immediate and informative feedback. This feedback enables continuous adjustment of behavior, creating a tight coupling between action and outcome. Rather than following fixed rules, individuals adapt in response to changing conditions. This aligns with the concept of recursive relevance realization, where cognition involves selecting what matters in a given context (Vervaeke et al., 2018). In flow, this selection process becomes highly refined, allowing individuals to focus on relevant information for the task at hand, while filtering out distractions.

The importance of this feedback loop can be illustrated through skilled activities such as sports or music. For example, a basketball player in flow does not consciously calculate each movement but adjusts their body, sometimes in midair, based on subtle cues including balance, timing, and the environment. These adjustments occur in real time, reflecting a continuous interaction between perception and action. This demonstrates that cognition in flow is not absent, but highly efficient and tightly integrated with the environment (Vervaeke et al., 2018; Seth & Hohwy, 2021).

The balance between challenge and skill further clarifies the function of consciousness in flow. Tasks that are too easy fail to generate sufficient feedback, leading to boredom and disengagement. Conversely, tasks that are too difficult produce excessive error, overwhelming cognitive resources and disrupting performance. Flow arises when the conditions of the task require one's existing skills to extend just beyond their current level, creating a state in which the individual is continuously challenged while still able to respond effectively. In this goldilocks zone, feedback remains informative and actionable, allowing for ongoing adjustment without destabilization. This suggests that consciousness functions as a system for maintaining optimal engagement with the environment by regulating the relationship between stability and adaptability (Seth & Hohwy, 2021; Gordon et al., 2017).

From this perspective, the function of consciousness is not merely to represent information, but to optimize action and learning in real time. It allows individuals to remain sensitive to relevant feedback while maintaining coherent, goal-oriented behavior. Flow represents an ideal expression of this function, where performance and learning are fully integrated (Vervaeke et al., 2018; Seth & Hohwy, 2021).

The Nature of Cognition in Flow: Insight and Self-Organization

Flow also provides insight into the nature of cognition itself. Vervaeke et al. (2018) emphasize that flow involves spontaneous insight, allowing individuals to restructure their understanding of a task. Unlike step-by-step reasoning, insight produces sudden reorganizations of the problem space. This suggests that cognition is not purely computational, but involves dynamic reconfiguration of internal representations.

Insight differs from both trial-and-error learning and automatic execution. In trial-and-error, improvement occurs gradually through repeated attempts, while in automaticity, behavior follows established patterns with minimal change. Insight, by contrast, involves a qualitative shift in how a problem is perceived. In flow, these shifts are not isolated events but occur continuously at a micro-level, allowing individuals to adapt fluidly as conditions change. This indicates that cognition in flow is characterized by ongoing restructuring rather than static processing. From a dynamical systems perspective, cognition is better understood as a continuously evolving process rather than a sequence of

discrete operations. Mental states are not fixed representations, but patterns that change in response to environmental input. Flow represents a state in which this adaptation occurs fluidly, without disruption. The individual is not explicitly analyzing each action, but responding directly to the unfolding structure of the task. This perspective challenges traditional computational models, which treat cognition as rule-based processing. While such models can explain structured problem-solving, they struggle to account for the flexibility and creativity observed in flow. Insight-based cognition demonstrates that the mind can reorganize itself in response to new information, rather than simply applying predefined rules.

Importantly, this suggests that the nature of consciousness is fundamentally dynamic and self-organizing. Consciousness is not a static representation of the world, but an active process that continuously reshapes itself in response to feedback. Flow makes this process especially visible, as it represents a state in which restructuring occurs rapidly and efficiently. Rather than being an exception to normal cognition, flow may reveal the underlying dynamics that are present in everyday experience but typically less apparent.

Competing Theories of Consciousness

Situating this account within broader debates helps clarify both its strengths and its limits. Several major theories offer different explanations of consciousness, including higher-order theories, workspace models, integrated information theory, and memory-based accounts. While each captures an important aspect of conscious experience, they tend to emphasize relatively static features and therefore struggle to account for the dynamic, adaptive processes revealed in flow. Higher-order theories propose that consciousness arises when mental states become the object of higher-order awareness (Brown et al., 2019). This highlights the role of reflection and metacognition, particularly in cases where individuals are explicitly aware of their own mental states. However, flow presents a challenge to this view. During flow, self-awareness is reduced, yet cognition remains highly adaptive and responsive. This suggests that higher-order representation is not necessary for all forms of conscious processing, especially those involving dynamic interaction with the environment.

Workspace theories offer a different perspective, proposing that consciousness emerges when information becomes globally available across cognitive systems (Luppi et al. 2021). These theories emphasize large-scale integration as a key feature, allowing perception, memory, and action to be coordinated. Flow can be partially understood within this framework, as it involves the seamless integration of multiple processes. However, workspace theories primarily focus on access and broadcasting, explaining how information becomes available rather than how it is transformed. As a result, they do not fully capture the role of insight and continuous restructuring that characterizes cognition in flow.

Integrated information theory (IIT) defines consciousness in terms of the quantity and structure of integrated information within a system (Negro, 2020). While this provides a formal account of complexity, it offers limited insight into function. Flow is not simply a state of high integration, but one in which cognition is actively reorganized in response to feedback. IIT may describe the richness of conscious experience, but it does not explain how that experience supports adaptive, real-time adjustment. Memory-based accounts propose that consciousness functions as a system for encoding, storing, and recombining experience (Budson et al., 2022). This perspective helps explain how past information informs present action, which is clearly relevant to skilled performance. However, flow involves more than the recombination of stored representations. It includes the generation of new

patterns through insight, suggesting that cognition is not only reconstructive but also continuously reorganizing itself in response to the environment.

Taken together, these theories highlight important dimensions of consciousness, including reflection, integration, structure, and memory. However, none fully account for the dynamic, self-organizing processes revealed in flow. Flow suggests that consciousness must be understood not as a static mechanism, but as an adaptive system capable of ongoing restructuring through the interaction of prediction, feedback, and relevance.

Predictive Processing and the Mechanisms of Flow

Predictive processing offers a unifying framework for understanding how flow arises by linking neural mechanisms to adaptive cognition. According to this view, the brain continuously generates predictions about the world and updates them based on sensory input (Seth & Hohwy, 2021). Cognition is therefore an inferential process aimed at minimizing prediction error, allowing the system to maintain a coherent and responsive model of its environment. In flow, this process appears to operate with a high degree of efficiency. Predictions are accurate enough to support smooth and coordinated action, while remaining flexible enough to accommodate change. This balance between stability and adaptability is essential. If predictions become too rigid, behavior loses flexibility and cannot adjust to new conditions. If they are too unstable, the system becomes overwhelmed by error and loses coherence. Therefore, flow can be understood as a state in which prediction and feedback are optimally calibrated, allowing for continuous adjustment without disruption.

Empirical evidence supports this interpretation, perception depends on the integration of top-down predictions and bottom-up sensory signals, and that this integration becomes more efficient as environmental predictability increases (Gordon et al. 2017). Neural markers of integration rise with certainty, indicating a stronger alignment between prediction and input. This suggests that flow may involve particularly effective coordination between these processes, where the system achieves a strong fit between expectation and environment. At the same time, predictive processing also helps explain the subjective qualities of flow. When prediction error is minimized, there is less need for conscious correction or deliberation, which can create the impression that actions are unfolding automatically. However, this does not imply the absence of cognition. Instead, it reflects a state in which cognitive processes are operating smoothly and efficiently, allowing the individual to remain fully engaged while continuously adapting to feedback.

Despite its strengths, predictive processing does not fully resolve the nature of consciousness on its own. As Seth and Hohwy (2021) note, it is better understood as a framework for consciousness science rather than a complete theory of consciousness. While it explains how prediction and error minimization support adaptive behavior, it does not fully account for why experience has the qualitative and meaningful structure that it does. In this respect, it shares limitations with other theories discussed earlier. Like workspace models, it emphasizes integration; like higher-order theories, it is associated with forms of awareness; and like memory-based accounts, it depends on stored information. However, none of these perspectives, taken in isolation, capture the full dynamics of insight-driven restructuring observed in flow.

Flow helps reveal both the strengths and limits of predictive processing. It supports the idea that cognition is fundamentally inferential and feedback-driven, but also shows that consciousness involves more than error minimization. It requires a system capable of dynamically reorganizing relevance in real time, integrating prediction, feedback, and insight into a unified process. Predictive processing provides the mechanistic foundation for this account, but flow demonstrates how these mechanisms are expressed in lived experience.

The Transcendental–Empirical Question

Understanding consciousness requires addressing both empirical mechanisms and the conditions that make experience possible. Empirically, predictive processing describes how neural systems generate and update predictions about the world through continuous interaction between top-down expectations and bottom-up sensory input (Seth & Hohwy, 2021). Studies such as Gordon et al. (2017) demonstrate that perception depends on the integration of these signals, and that this integration becomes more efficient as environmental predictability increases. These findings provide a measurable account of how the brain supports adaptive perception and action.

However, empirical accounts alone do not explain why experience has the structure that it does. The transcendental question concerns what must be true for meaningful experience to occur at all. Flow suggests that experience requires more than neural activity, it requires a system capable of selecting what is relevant and reorganizing itself in response to feedback. This aligns with the idea that cognition is fundamentally shaped by relevance realization, where the mind continuously prioritizes certain aspects of the environment over others (Vervaeke et al., 2018). Without this capacity, perception would either be overwhelmed by irrelevant information or reduced to rigid, unadaptive patterns.

Flow helps bridge these two levels. Empirically, it reflects efficient neural integration and reduced prediction error, as described by predictive processing frameworks (Seth & Hohwy, 2021; Gordon et al., 2017). Transcendentally, it reveals that consciousness requires a system capable of dynamically organizing relevance in real time. In flow, this organization is especially clear, attention is tightly coupled to action, irrelevant information is suppressed, and feedback is continuously incorporated into behavior. This suggests that consciousness is not simply the product of neural processes, but the structured relationship between those processes and meaningful engagement with the environment. Therefore, consciousness cannot be fully reduced to neural activity alone, instead it must be understood as arising from the interaction between underlying mechanisms and the conditions that allow experience to be coherent, adaptive, and meaningful. Flow provides a concrete example of this interaction, showing how empirical processes and transcendental structure come together in a unified form of cognition.

The Mindedness Question

A final question concerns what distinguishes a minded system from a non-minded one. Simple information processing is not sufficient, as many artificial and biological systems can process information without exhibiting the flexibility and meaningful engagement associated with minds. What distinguishes minded systems, on the account developed here, is their capacity for adaptive, self-organizing interaction with the environment.

Flow provides a clear illustration of these features. Individuals in flow are not executing fixed routines, but continuously adjusting their behavior in response to changing conditions and ongoing feedback. This requires more than the passive processing of information, it involves the active evaluation of what is relevant in a given context and the ability to reorganize internal models accordingly. In this sense, cognition in flow is both selective and transformative, as the system is constantly refining its understanding of the task in real time. This aligns with the concept of relevance realization, where cognition is guided by the dynamic prioritization of information based on its significance for action (Vervaeke et al., 2018). Predictive processing supports this view by framing cognition as an ongoing process of inference and model updating (Seth & Hohwy, 2021). A minded system does not simply react to stimuli, but actively generates predictions about the environment and adjusts those predictions in response to error signals. This capacity allows for flexible and context-sensitive behavior, distinguishing minded systems from those that operate through fixed or pre-programmed responses. In flow, this process becomes especially efficient, as prediction and feedback are tightly coupled, enabling rapid and continuous adaptation.

A mind, therefore, is best understood as a system capable of dynamic relevance realization and continuous restructuring. While reflection, representation, and information integration may contribute to mindedness, they are not sufficient on their own. What is essential is the ability to adapt in real time through the integration of prediction, feedback, and relevance. Flow represents this capacity in its most refined form, making it a useful model for understanding what it means to be a minded system.

Objections and Responses

One objection is that flow can be explained entirely through automaticity. According to this view, flow occurs when skills become so well learned that they no longer require conscious control. While this explanation captures the smoothness of performance, it fails to account for the adaptive and creative features of flow. Automatic processes are typically rigid, whereas flow involves continuous adjustment and improvement. The presence of insight and learning suggests that flow cannot be reduced to unconscious execution.

A second objection is that flow represents a non-ordinary state of consciousness and therefore provides limited insight into everyday cognition. On this view, flow is too exceptional to serve as a general model of consciousness. However, this objection underestimates the value of studying extreme or optimized states. Flow may exaggerate processes that are present in everyday cognition, such as prediction, feedback integration, and relevance selection. By making these processes more visible, flow allows us to better understand how consciousness operates more generally.

A third objection is that predictive processing is too broad to provide a specific explanation of consciousness. Because it applies to many aspects of cognition, it may seem insufficiently precise. However, as Seth and Hohwy (2021) argue, the goal is not to reduce consciousness to a single mechanism, but to develop systematic mappings between neural processes and phenomenological experience. Predictive processing achieves this by linking biological mechanisms to the structure of experience, offering a flexible but coherent framework.

Conclusion

Flow is commonly understood as a state of effortless performance in which cognition appears to recede. However, this paper has argued that such interpretations are misleading. Drawing on Vervaeke et al. (2018) and predictive processing frameworks, flow is better understood as a state in which cognition is not diminished, but reorganized. Rather than reflecting the suppression of higher processes, flow reveals how insight, prediction, and feedback can be integrated to support continuous adaptation in real time.

As a case study, flow provides a unique window into the structure of consciousness. It sits at the intersection of skilled action, subjective experience, and cognitive transformation, making visible processes that are often less apparent in everyday cognition. Across the four central questions addressed in this essay, a consistent picture emerges. Functionally, consciousness supports the optimization of action and learning through relevance realization. In terms of its nature, it is not static or purely computational, but dynamic and self-organizing. At the transcendental–empirical level, consciousness reflects both underlying neural mechanisms and the conditions required for meaningful experience. Finally, with respect to mindedness, flow demonstrates that a mind is defined by its capacity for adaptive, real-time restructuring in response to feedback. Taken together, these points suggest that consciousness is best understood not as a passive or representational system, but as an active process that continuously reshapes itself in relation to the environment. Flow does not represent an exception to normal cognition, but rather an optimized expression of it. Understanding flow therefore provides insight into the broader nature of consciousness, highlighting its role in adaptation, integration, and transformation.

References

- Brown, R., Lau, H., & LeDoux, J. E. (2019). Understanding the higher-order approach to consciousness. *Trends in Cognitive Sciences*, 23(9), 754–768. <https://doi.org/10.1016/j.tics.2019.06.009>
- Budson, A. E., Richman, K. A., & Kensinger, E. A. (2022). Consciousness as a memory system. *Cognitive and Behavioral Neurology*, 35(1), 1–12. <https://doi.org/10.1097/WNN.0000000000000300>
- Gordon, N., Koenig-Robert, R., Tsuchiya, N., van Boxtel, J. J. A., & Hohwy, J. (2017). Neural markers of predictive coding under perceptual uncertainty revealed with hierarchical frequency tagging. *eLife*, 6, e22749. <https://doi.org/10.7554/eLife.22749>
- Luppi, A. I., Mediano, P. A. M., Rosas, F. E., Harrison, D. J., Carhart-Harris, R. L., Bor, D., Stamatakis, E. A., & Seth, A. K. (2021). A synergistic workspace for human consciousness revealed by integrated information decomposition. *Current Biology*, 31(13), 2883–2894.e3. <https://doi.org/10.1016/j.cub.2021.04.046>
- Negro, N. (2020). Phenomenology-first versus third-person approaches in the science of consciousness: The case of integrated information theory and the unfolding argument. *Phenomenology and the Cognitive Sciences*, 19(5), 979–996. <https://doi.org/10.1007/s11097-020-09681-3>

Seth, A. K., & Hohwy, J. (2021). Predictive processing as an empirical theory for consciousness science. *Cognitive Neuroscience*, 12(2), 89–90. <https://doi.org/10.1080/17588928.2020.1838467>

Vervaeke, J., Ferraro, L., & Herrera-Bennett, A. (2018). Flow as spontaneous thought: Insight and implicit learning in skilled performance. In K. C. R. Fox & K. Christoff (Eds.), *The Oxford handbook of spontaneous thought: Mind-wandering, creativity, and dreaming* (pp. 333–346). Oxford University Press.